WORKSHOP · DAY 01 · CHEAT SHEET

LLMs for political scientists. What kind do you want?

WORKSHOP MATERIALS
GITHUB.COM/JACQPARK/
INSTATS-LLM-WORKSHOP

THE TWO FAMILIES

Encoder

READ. LABEL. DONE.

- BERT, RoBERTa, DeBERTa, ModernBERT
- Reads a passage in full, returns a fixed prediction (label, score)
- Deterministic, cheap, fast
- Runs on a laptop or free Colab CPU
- Good for stance, sentiment, topic, NER, any closed-label task

Generative

CONTINUE THE TEXT.

- Claude, GPT, Gemini, LLaMA, Qwen
- Given a prompt, produces more text
- Stochastic, costly per call, slower
- Usually called via API
- Good for extraction, summarization, free-form judgment
- Reach for it when the answer cannot fit in a fixed label set

DECISION · 30-SECOND VERSION

	CLOSED LABEL SET	FREE-FORM OUTPUT	MIXED
Small data ~hundreds to thousands	**ENCODER, ZERO-SHOT** Laurer DeBERTa-v3 with your own labels. No training.	**GENERATIVE, FEW-SHOT** Claude or Gemini with 3 to 5 examples in the prompt.	**GENERATIVE, PROMPTED** LLM with clear schema and validators.
Large data ~tens of thousands+	**ENCODER, FINE-TUNED** 300 hand-labels, train an encoder.	**ENCODER + GENERATIVE** Encoder filter, generative for hard cases.	**HYBRID PIPELINE** Generative for extraction, encoder to classify.

COST · SPEED · REPRODUCIBILITY

AXIS	ENCODER	GENERATIVE
Cost / 1k texts	~$0 after model download	A few cents to a few dollars (per token)
Speed	~50 to 500 texts / sec on free GPU	~1 to 5 calls / sec over network
Determinism	Same input, same output, every time	Stochastic; pin temperature 0 + version
Validation	k-fold CV, confusion matrix, kappa	Spot-check, schema validation, audits
Replication	Pin model ID, ship weights with code	Pin version; log every prompt + response

THE NLI TRICK · WHY AN ENCODER CAN DO ALMOST ANYTHING

An NLI-trained encoder learns one job. Given a **premise** and a **hypothesis**, decide entailment / neutral / contradiction. Any classification can be rewritten as a yes-or-no entailment question. `zero-shot-classification` does this for you with a list of candidate labels. Park (2026) aggregates 13 such hypotheses with weights into a continuous -5 to +5 stance score.

CODE · TWO SNIPPETS TO KEEP

ENCODER · ZERO-SHOT CLASSIFICATION

```python
# pip install -q transformers
from transformers import pipeline

clf = pipeline(
  "zero-shot-classification",
  model="MoritzLaurer/DeBERTa-v3-"
        "base-mnli-fever-anli",
)
clf(text, candidate_labels=[
  "weak IP protection",
  "strong IP protection",
```

GENERATIVE · STRUCTURED EXTRACTION

```python
# pip install -q anthropic
from anthropic import Anthropic
import json

client = Anthropic()
resp = client.messages.create(
  model="claude-haiku-4-5-20251001",
  max_tokens=300,
  messages=[{"role": "user",
    "content": f"Return JSON: {schema}"}],
)
data = json.loads(resp.content[0].text)
```

```
  "unrelated",
])
```

MODELS PICKED FOR THIS WORKSHOP

`MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli`
Encoder · zero-shot via NLI · ~735 MB · MIT · primary measurement workhorse

`cardiffnlp/twitter-roberta-base-sentiment-latest`
Encoder · 3-class sentiment · ~500 MB · CC-BY-4.0 · sentiment baseline

`claude-haiku-4-5-20251001`
Generative · cheap and fast · Anthropic API · Day 3 extraction tasks

`Qwen/Qwen2.5-1.5B-Instruct`
Generative · open weights · runs on free Colab GPU · no-key fallback

PITFALLS

- **"Looks impressive" ≠ publishable.** Always validate against a hand-coded gold set.
- **Off-the-shelf trips on interpretive text.** Mixed praise + criticism needs hand-labels.
- **Wording of candidate labels matters.** The model reads them as hypotheses, not keywords.
- **Pin model versions.** Cloud APIs retire models. HF model IDs survive.
- **Never paste API keys in code cells.** Use Colab's `userdata` sidebar.